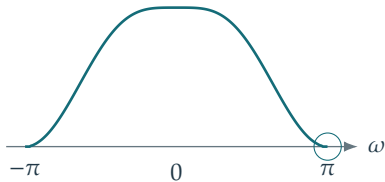


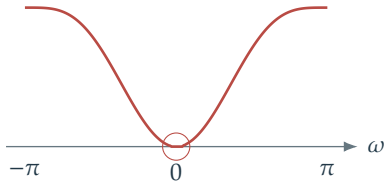
two vanishing moments illustrated ($p = 2$)

$|\widehat{h}(\omega)|$



order- p zero at π

$|\widehat{g}(\omega)|$



order- p zero at 0

$$\widehat{h}^{(k)}(\pi) = \widehat{g}^{(k)}(0) = 0, \quad k = 0, \dots, p-1$$

$$\iff \int_{\mathbb{R}} x^k \psi(x) \, dx = 0, \quad k = 0, \dots, p-1$$

Curves show magnitudes; derivatives refer to the filter symbols.